

Time to Treatment in Stroke Thrombectomy and Outcomes in the Extended Time Window

A Meta-Analysis

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Abstract

Background and Objectives

Time to endovascular thrombectomy (EVT) is a critical determinant of outcomes for large vessel occlusion (LVO) strokes presenting within 6 hours of onset. Its impact in the extended (6–24-hour) window remains uncertain. We aimed to evaluate the association between treatment times and outcomes in this window.

Methods

Individual patient-level data from 6 randomized trials enrolling patients in the extended window were pooled. The primary outcome was degree of disability at 90 days (modified Rankin Scale [mRS] 0–6). Secondary outcomes included functional independence (mRS 0–2), mortality, and symptomatic intracranial hemorrhage.

Results

Among 505 participants (median age 70 years; baseline NIH Stroke Scale 16; 51.3% female; advanced imaging selection 86.5%), 266 (52.7%) received EVT and 239 (47.3%) medical therapy alone. In EVT-treated patients, longer onset-to-randomization times were not associated with differences in disability (adjusted odds ratio [aOR] per 60 minutes 1.02; 95% CI 0.96–1.08; $p = 0.53$) or functional independence (aOR 1.06; 95% CI 0.95–1.17; $p = 0.28$). Conversely, control patients exhibited worse outcomes with increasing onset-to-randomization times (aOR for disability 0.93; 95% CI 0.87–1.00; $p = 0.041$; functional independence 0.84; 95% CI 0.73–0.97; $p = 0.019$), resulting in greater treatment benefit at later times (p -interaction = 0.033 and 0.003, respectively). No association was observed between onset-to-puncture or onset-to-reperfusion and outcomes in EVT patients. However, longer randomization-to-reperfusion times correlated with worse disability (aOR 0.58; 95% CI 0.37–0.91; $p = 0.018$) and lower functional independence (aOR 0.51; 95% CI 0.28–0.96; $p = 0.038$).

Discussion

This study demonstrates that the benefit of EVT is consistently preserved across the extended (6–24-hour) time window when patients are selected using advanced imaging criteria. In this highly selected population, enriched with slow progressors, the apparent increase in treatment effect with longer onset-to-presentation times reflects the validity of physiologic selection rather than true time insensitivity. However, this observation should not diminish the critical importance of time: imaging-based selection effectively resets the treatment clock, masking the harmful impact of prehospital delays. Notably, longer in-hospital delays were strongly associated with worse outcomes, reinforcing that “time is brain,” even in extended-window cohorts.

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Glossary

aOR = adjusted odds ratio; **cOR** = common odds ratio; **ED** = emergency department; **EVT** = endovascular thrombectomy; **IQR** = interquartile range; **LVO** = large vessel occlusion; **mRS** = modified Rankin Scale; **mTICI** = modified Thrombolysis in Cerebral Infarction; **NIHSS** = NIH Stroke Scale; **RCT** = randomized clinical trial; **sICH** = symptomatic intracranial hemorrhage; **VIF** = variance inflation factor.

Introduction

Several landmark randomized clinical trials (RCTs) have established endovascular thrombectomy (EVT) as the standard of care for acute ischemic stroke due to large vessel occlusions (LVOs) in the anterior circulation, in both early (0–6 hours) and extended (6–24 hours from stroke onset) time windows.^{1,2} The strong association between time to EVT and clinical outcomes in the early window is well documented.^{3–9} However, the effect of time to EVT on outcomes in the extended window remains less well understood.

The AURORA collaboration pooled individual patient-level data from 6 modern EVT RCTs enrolling anterior circulation LVO patients with small to moderate infarct sizes randomized in the 6–24-hour window.^{10–15} In this study, we aimed to evaluate the influence of various time metrics on outcomes in the AURORA (Analysis of Pooled Data from Randomized StUdies of ThROMbectomy MoRe than 6 hours After Last Known Well) meta-analysis to provide more precise data about the impact of time to EVT on clinical outcomes in the extended time window. In addition, we explored the potential mechanisms underlying these associations.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

The individual randomized controlled trials included in this meta-analysis were each approved by the Institutional Review Boards or Ethics Committees at every participating site. Written informed consent was obtained from all participants or their legally authorized representatives for participation in the original trials and for the secondary use of deidentified data. No recognizable persons or personal identifiers are included in the manuscript; therefore, additional consent to disclose was not required.

Study Design and Outcomes

AURORA is a pooled analysis of 6 published randomized trials (DAWN, DEFUSE 3, ESCAPE, REVASCAT, POSITIVE, and RESILIENT)^{10–15} that included patients with anterior circulation LVO patients with clinical imaging (characterized by moderate to severe clinical deficits and small to moderate infarct sizes) who were randomized beyond 6 hours from time last known well (TLKW) considered by convention to represent the time of stroke onset and were treated with second-generation neuro-thrombectomy devices.

All trials were conducted with ethics approval from the local institutional review boards at participating sites.

We analyzed data to explore the association between different time metrics and outcomes. The time metrics of interest included time from stroke onset-to-randomization in both EVT and control patients as well as time from stroke onset-to-arterial puncture, time from stroke onset-to-reperfusion, and time from randomization-to-reperfusion in the EVT patients. We also characterized the patients according to time from stroke onset-to-randomization into 3-time windows (6–12 hours, 12–16 hours, and 16–24 hours) to better explore any differences in baseline characteristics according to presentation time. As in the original AURORA analysis, the prespecified primary outcome was the degree of disability on the modified Rankin Scale (mRS) at 90 days, which ranges from normal (0) to death (6), with scores of 5 (severe disability) and 6 collapsed into 1 category. The secondary efficacy outcome was the rate of functional independence (mRS 0–2) at 90 days. Safety outcomes included 90-day mortality and periprocedural symptomatic intracranial hemorrhage (sICH) according to the definition used in each trial. Successful reperfusion was defined as achieving a modified Thrombolysis in Cerebral Infarction (mTICI) scale score of 2b or 3 (corresponding to reperfusion of at least 50% of the affected vascular territory) as adjudicated by the core laboratory of each individual trial.¹⁶

Statistical Analysis

The Study Protocol and Statistical Analysis Plan is available in the eSAP. Time-dependent outcome probabilities were estimated using mixed-effects ordinal logistic regression for the mRS distribution and mixed-effects binary logistic regression for dichotomous end points. Trial and trial-by-treatment interaction were modeled as random effects to account for between-study variability. Primary models evaluated the linear relationship between the log odds of each outcome and treatment allocation (endovascular vs medical management), with time entered as a continuous linear covariate. In combined analyses incorporating both treatment arms, a time-by-treatment interaction term was included to test whether the effect of thrombectomy varied across time. The primary effect estimate was the common odds ratio (cOR), representing the odds of achieving a more favorable position on the mRS ordinal scale. The absolute risk difference, defined as the proportion of patients achieving improvement by at least 1 mRS category, was estimated by averaging predictions from the algorithmic joint outcome table and permutation-based approaches.

For dichotomous outcomes, absolute risk differences were derived from the difference in model-predicted probabilities. All adjusted models included age (continuous) and baseline NIH Stroke Scale (NIHSS) score as covariates.

Data acquisition, harmonization, and quality assurance procedures for the AURORA collaboration have been described in detail previously. A standardized minimum data set was specified by the collaborative investigators, extracted independently by each trial's statistician, and submitted to a central independent analyst for merging and verification. Harmonization of variable definitions and resolution of discrepancies were performed in coordination with each trial's data center. Validation included comparison of independently derived results against published trial-level findings and unpublished summaries. All analyses were conducted on the final harmonized data set.

Four-time intervals were examined: (1) symptom onset-to-randomization, (2) randomization-to-reperfusion, (3) symptom onset-to-expected arterial puncture, and (4) symptom onset-to-expected reperfusion. Symptom onset was defined as the TLKW. Expected arterial puncture and expected reperfusion times were imputed for all patients, including those in the medical therapy arm, by adding the trial-specific mean randomization-to-puncture or randomization-to-reperfusion interval to each patient's individual onset-to-randomization time. In a separate analysis restricted to endovascular patients who achieved substantial reperfusion (mTICI score of 2b or 3), the primary time variable was the actual onset-to-reperfusion interval.

Collinearity was assessed using variance inflation factors (VIFs) for all covariates in multivariable models; all VIF values were <2.0 for age and baseline NIHSS. Time metrics were analyzed in separate models rather than simultaneously to avoid the inherent collinearity between overlapping time intervals (e.g., onset-to-randomization and onset-to-puncture). All effect estimates are reported with accompanying 95% confidence intervals. Two-sided *p* values below 0.05 were considered statistically significant; no corrections for multiple comparisons were applied. Analyses were conducted in SAS version 9.4 (SAS Institute, Cary, NC), and figures were generated using R version 4.2 (R Foundation for Statistical Computing, Vienna, Austria).

Data Availability

The individual deidentified participant data used in this individual patient data meta-analysis are maintained at the AURORA statistical core laboratory. Proposals for the use of data from this study should be directed to S. Brown. To gain access, data requestors will need to sign a data-sharing agreement and provide a statistically sound research proposal that is approved by the AURORA Steering Committee. Related documents, including the study protocol and statistical analysis plan, will be made available upon reasonable request to the corresponding author. Data will be available to qualified researchers for the purpose of replicating the results or conducting further meta-

analyses, provided that such use does not conflict with the primary trial data use policies or ethical constraints.

Results

Of the 505 participants from the 6 included trials, 266 (52.7%) were assigned to the EVT arm and 239 (47.3%) were assigned to the control arm. The comparison of baseline characteristics and outcomes across the 3-time windows (6–12 hours, 12–16 hours, and 16–24 hours) is summarized in Table 1. Overall, the mean (SD) of age was 68.6 years (± 13.7) with 259 (51.3%) participants being women. The median (interquartile range [IQR]) time from onset to randomization was 625 minutes (472–808). The median (IQR) baseline NIHSS was 16 (13–20), and the median (IQR) ASPECTS was 8 (7–9). As expected, the rate of intravenous alteplase use declined over time (6–12 hours: 18.9%; 12–16 hours: 2.5%; 16–24 hours: 0%; $p < 0.001$). Moreover, there was a significant difference in the mode of onset across the 3-time epochs with a shift in proportion of witnessed to unwitnessed patients over time ($p < 0.001$) (Table 1). This was most pronounced in the 16–24-hour interval (only DAWN trial patients). The distribution of participants by trial across time epochs is presented in eTable 1. The 6–12-hour window included participants from all 6 trials. The 12–16-hour window was contributed equally by DAWN and DEFUSE-3 (50% each). The 16–24-hour window comprised exclusively DAWN trial patients, as no other included trial enrolled patients beyond 16 hours from onset. The relationships between the different time metrics and outcomes of interest are presented in Table 2.

Association Between Time From Stroke Onset to Randomization and Outcomes

In the EVT group, longer stroke onset-to-randomization times were not associated with significant differences in overall disability (adjusted odds ratio [aOR] for achieving a lower disability status per 60 minutes 1.02, 95% CI 0.96–1.08; $p = 0.53$) or functional independence (aOR per 60 minutes 1.06, 95% CI 0.95–1.17; $p = 0.28$) at 90 days. By contrast, control patients showed a significant decrease in the odds of achieving a lower disability status (aOR per 60 minutes 0.93, 95% CI 0.87–1.00; $p = 0.041$) and functional independence (aOR per 60 minutes 0.84, 95% CI 0.73–0.97; $p = 0.019$) with longer onset-to-randomization times, which explains the significantly higher treatment effects of EVT with longer onset-to-randomization times (p -interaction = 0.033 and 0.003, respectively) (Figure 1A).

Patients with EVT exhibited a nonsignificant decrease in 90-day mortality over time (aOR per 60 minutes 0.93, 95% CI 0.84–1.03; $p = 0.15$), while control patients showed a nonsignificant increase (aOR per 60 minutes 1.08, 95% CI 0.97–1.19; $p = 0.15$), with a significantly higher treatment effect of EVT in mortality over time (p -interaction = 0.031) (Figure 1B). The mortality analysis (Figure 1B) showed wide

Table 1 Baseline Characteristics and Outcomes According to Time From Stroke Onset to Randomization

Variable	6–12-h			12–16-h			16–24-h		
	Overall	EVT	Control	Overall	EVT	Control	Overall	EVT	Control
Age (y)									
Mean ± SD (N)	67.2 ± 14.1 (333)	66.9 ± 14.4 (177)	67.5 ± 13.7 (156)	70.7 ± 12.8 (120)	71.8 ± 11.1 (61)	69.6 ± 14.3 (59)	72.2 ± 12.5 (52)	70.4 ± 13.7 (28)	74.4 ± 10.9 (24)
Median (range)	68 (58–78.2)	68 (57–78)	68.6 (58–79)	73 (61–80)	73 (63–79)	73 (58.5–80.5)	75 (63–82)	74.5 (62–79)	77.5 (64–83)
Female, % (n/N)	47.1 (157/333)	50.3 (89/177)	43.6 (68/156)	57.5 (69/120)	65.6 (40/61)	49.2 (29/59)	63.5 (33/52)	60.7 (17/28)	66.7 (16/24)
Baseline NIHSS									
Mean ± SD (N)	16.5 ± 5.4 (332)	16.1 ± 4.8 (176)	17.0 ± 6.0 (156)	16.7 ± 5.9 (120)	16.2 ± 5.9 (61)	17.3 ± 5.8 (59)	18.6 ± 5.5 (52)	18.9 ± 6.2 (28)	18.2 ± 4.6 (24)
Median (range)	16 (12–20)	15.5 (13–20)	16 (12–21.3)	16.5 (12–21)	16 (11–21)	17 (14–20)	18 (15–24)	17.5 (13–25)	18 (15–20.8)
Baseline ASPECTS									
Mean ± SD (N)	7.7 ± 1.4 (291)	7.8 ± 1.2 (157)	7.6 ± 1.6 (134)	7.7 ± 1.4 (113)	7.9 ± 1.4 (60)	7.5 ± 1.4 (53)	7.0 ± 1.7 (51)	7.3 ± 1.6 (28)	6.7 ± 1.7 (23)
Median (range)	8 (7–9)	8 (7–9)	8 (7–9)	8 (7–9)	8 (7–9)	7 (7–9)	8 (6–8)	8 (6–8)	7 (5.5–8)
Atrial fibrillation, % (n/N)	34.2 (104/304)	37.5 (60/160)	30.6 (44/144)	37.0 (44/119)	45.0 (27/60)	28.8 (17/59)	30.0 (15/50)	33.3 (9/27)	26.1 (6/23)
Hypertension	74.4 (247/332)	69.3 (122/176)	80.1 (125/156)	76.5 (91/119)	83.3 (50/60)	69.5 (41/59)	76.9 (40/52)	82.1 (23/28)	70.8 (17/24)
Hyperlipidemia	47.1 (153/325)	50.3 (87/173)	43.4 (66/152)	56.3 (67/119)	63.3 (38/60)	49.2 (29/59)	54.0 (27/50)	48.1 (13/27)	60.9 (14/23)
Diabetes mellitus	27.6 (91/330)	25.7 (45/175)	29.7 (46/155)	26.9 (32/119)	30.0 (18/60)	23.7 (14/59)	25.5 (13/51)	18.5 (5/27)	33.3 (8/24)
Glucose (mg/dL)									
Mean ± SD (N)	139.3 ± 50.8 (327)	138.4 ± 49.5 (176)	140.4 ± 52.4 (151)	133.3 ± 47.1 (120)	132.5 ± 48.1 (61)	134.1 ± 46.4 (59)	125.6 ± 42.2 (52)	113.5 ± 21.9 (28)	139.8 ± 54.7 (24)
Median (range)	125 (108.7–152)	124.1 (108.8–150.3)	126 (108.5–152.5)	118 (106.0–140.5)	119 (107–135)	118 (106–149.5)	113 (103.3–132.8)	113 (101–125.3)	118 (106.8–152.5)
tPA administered	18.9 (63/333)	15.3 (27/177)	23.1 (36/156)	2.5 (3/120)	1.6 (1/61)	3.4 (2/59)	0 (0/52)	0 (0/28)	0 (0/24)
Systolic blood pressure (mm Hg)									
Mean ± SD (N)	145.0 ± 22.1 (333)	145.4 ± 21.6 (177)	144.5 ± 22.7 (156)	147.3 ± 20.8 (120)	145.9 ± 21.0 (61)	148.8 ± 20.7 (59)	153.0 ± 23.7 (52)	152.8 ± 26.1 (28)	153.3 ± 21.2 (24)
Median (range)	144 (130–159)	144 (132–160)	143 (129.9–158.3)	149 (132.8–160.3)	147 (132–157)	151 (134.5–162.5)	152 (140.8–168.5)	152 (140.8–165.8)	151 (139.5–170)
Intracranial occlusion location, n% (n/N)									

Continued

Table 1 Baseline Characteristics and Outcomes According to Time From Stroke Onset to Randomization (*continued*)

Variable	6–12-h			12–16-h			16–24-h		
	Overall	EVT	Control	Overall	EVT	Control	Overall	EVT	Control
Internal carotid artery	22.2 (74/333)	23.2 (41/177)	21.2 (33/156)	35.8 (43/120)	31.1 (19/61)	40.7 (24/59)	21.2 (11/52)	21.4 (6/28)	20.8 (5/24)
M1 segment middle cerebral artery	74.2 (247/333)	72.9 (129/177)	75.6 (118/156)	62.5 (75/120)	68.9 (42/61)	55.9 (33/59)	76.9 (40/52)	78.6 (22/28)	75.0 (18/24)
M2 segment middle cerebral artery	3.6 (12/333)	4.0 (7/177)	3.2 (5/156)	1.7 (2/120)	0.0 (0/61)	3.4 (2/59)	1.9 (1/52)	0.0 (0/28)	4.2 (1/24)
Process times, min, mean ± SD (N); median (range)									
Onset to randomization	527.4 ± 106.2 (333) 524.0 (427.9–623.7)	529.7 ± 111.2 (177) 531.3 (422–630.6)	524.7 ± 100.4 (156) 517.9 (434.5–609)	840.7 ± 72.7 (120) 840.0 (781.5–905.1)	832.9 ± 82.9 (61) 835.4 (743–908)	848.6 ± 60.0 (59) 850 (800.7–896.8)	1,136.4 ± 129.6 (52) 1,099 (1,031–1,216.3)	1,133.5 ± 115.8 (28) 1,098 (1,053.4–1,199.2)	1,139.7 ± 146.6 (24) 1,099.7 (1,012.3–1,247.3)
Onset to puncture	N/A	560.7 ± 112.6 (176) 555 (455–659)	N/A	N/A	856.9 ± 82.9 (61) 851 (775–940)	N/A	N/A	1,156.0 ± 112.9 (28) 1,115 (1,074.8–1,213.8)	N/A
Onset to reperfusion	N/A	616.1 ± 112.0 (127) 627 (510.5–706.5)	N/A	N/A	913.8 ± 91.9 (51) 923 (822.5–990)	N/A	N/A	1,227.6 ± 126.0 (23) 1,183 (1,128.5–1,299)	N/A
Mode of onset, n% (n/N)									
Witnessed	31.0 (67/216)	28.2 (31/110)	34.0 (36/106)	18.3 (22/120)	18.0 (11/61)	18.6 (11/59)	3.8 (2/52)	0.0 (0/28)	8.3 (2/24)
Wakeup	48.1 (104/216)	57.3 (63/110)	38.7 (41/106)	60.8 (73/120)	60.7 (37/61)	61.0 (36/59)	53.8 (28/52)	57.1 (16/28)	50.0 (12/24)
Unwitnessed	20.8 (45/216)	14.5 (16/110)	27.4 (29/106)	20.8 (25/120)	21.3 (13/61)	20.3 (12/59)	42.3 (22/52)	42.9 (12/28)	41.7 (10/24)
Outcomes									
Median 90-d mRS	3.2 ± 1.9 (332) 3 (2–5)	3.0 ± 2.0 (177) 3 (1–5)	3.5 ± 1.8 (155) 4 (2–5)	3.6 ± 1.8 (120) 4 (2–5)	2.9 ± 1.8 (61) 3 (1–4)	4.4 ± 1.5 (59) 4 (3.5–6)	3.7 ± 1.7 (52) 4 (3–5)	3.3 ± 1.8 (28) 3 (2–4.3)	4.3 ± 1.3 (24) 4 (3.8–5.3)
90-d mRS 0–2	37.3 (124/332)	47.5 (84/177)	25.8 (40/155)	27.5 (33/120)	45.9 (28/61)	8.5 (5/59)	21.2 (11/52)	35.7 (10/28)	4.2 (1/24)
Successful reperfusion (mTICI 2b–3)	80.7 (142/176)	80.6 (141/175)	100.0 (1/1)	81.7 (49/60)	81.7 (49/60)		82.1 (23/28)	82.1 (23/28)	
sICH	4.2 (14/333)	5.1 (9/177)	3.2 (5/156)	5.8 (7/120)	6.6 (4/61)	5.1 (3/59)	1.9 (1/52)	3.6 (1/28)	0.0 (0/24)
90-d mortality	16.6 (55/332)	18.1 (32/177)	14.8 (23/155)	20.0 (24/120)	11.5 (7/61)	28.8 (17/59)	21.2 (11/52)	17.9 (5/28)	25.0 (6/24)

Abbreviations: ASPECTS = Alberta Stroke Program Early CT Score; EVT = endovascular thrombectomy; mRS = modified Rankin Scale; mTICI = modified Thrombolysis in Cerebral Infarction; NIHSS = NIH Stroke Scale; sICH = symptomatic intracranial hemorrhage; TPA = tissue plasminogen activator.

Table 2 Association Between Time Metrics and Outcomes in EVT and Control Groups

Time metric	Outcome	EVT		Control		Interaction <i>p</i> value
		OR (95% CI) per 1-h delay	ARD (%)	OR (95% CI) per 1-h delay	ARD (%)	
A. Onset to randomization	mRS shift at 90 d	1.02 (0.96–1.08), <i>p</i> = 0.53	NA	0.93 (0.87–1.00), <i>p</i> = 0.041	NA	0.033
	mRS 0–2 at 90 d	1.06 (0.95–1.17), <i>p</i> = 0.28	2.0	0.84 (0.73–0.97), <i>p</i> = 0.019	–4.8	0.003
	Mortality at 90 d	0.93 (0.84–1.03), <i>p</i> = 0.15	–0.6	1.08 (0.97–1.19), <i>p</i> = 0.15	0.8	0.031
	sICH at 24 h	0.98 (0.84–1.14), <i>p</i> = 0.76	0.1	0.94 (0.78–1.09), <i>p</i> = 0.70	–0.2	0.74
B. Randomization to reperfusion	mRS shift at 90 d	0.58 (0.37–0.91), <i>p</i> = 0.018	NA	NA	NA	NA
	mRS 0–2 at 90 d	0.51 (0.28–0.96), <i>p</i> = 0.038	–12.2	NA	NA	NA
	Mortality at 90 d	1.28 (0.58–2.82), <i>p</i> = 0.55	4.5	NA	NA	NA
	sICH at 24 h	1.12 (0.41–3.07), <i>p</i> = 0.82	0.7	NA	NA	NA
C. Onset to arterial puncture	mRS shift at 90 d	1.01 (0.96–1.07), <i>p</i> = 0.66	NA	0.94 (0.88–1.00), <i>p</i> = 0.058	NA	0.093
	mRS 0–2 at 90 d	1.04 (0.95–1.13), <i>p</i> = 0.41	1.9	0.84 (0.73–0.98), <i>p</i> = 0.022	–4.7	0.014
	Mortality at 90 d	0.94 (0.85–1.04), <i>p</i> = 0.24	–0.3	1.09 (0.99–1.20), <i>p</i> = 0.094	1.0	0.037
	sICH at 24 h	0.96 (0.82–1.13), <i>p</i> = 0.65	–0.1	0.96 (0.79–1.17), <i>p</i> = 0.69	–0.1	0.82
D. Onset to reperfusion	mRS shift at 90 d	0.99 (0.93–1.06), <i>p</i> = 0.80	NA	0.94 (0.88–1.00), <i>p</i> = 0.052	NA	0.27
	mRS 0–2 at 90 d	0.99 (0.90–1.10), <i>p</i> = 0.91	–0.2	0.85 (0.73–0.97), <i>p</i> = 0.020	–4.5	0.086
	Mortality at 90 d	0.93 (0.82–1.06), <i>p</i> = 0.28	–0.4	1.09 (0.99–1.19), <i>p</i> = 0.095	1.0	0.049
	sICH at 24 h	1.03 (0.89–1.20), <i>p</i> = 0.70	0.1	0.96 (0.79–1.17), <i>p</i> = 0.70	–0.1	0.58

Abbreviations: ARD = absolute risk difference; EVT = endovascular thrombectomy; mRS = modified Rankin Scale; OR = odds ratio; sICH = symptomatic intracranial hemorrhage.

and overlapping confidence intervals reflecting the relatively low event rate, and the results should be interpreted with appropriate caution despite the significant interaction term. There was no association between stroke onset-to-randomization time and sICH in either EVT or control patients (eFigure 1).

Association Between Treatment Times and Outcomes

Significantly higher treatment effects were observed with longer onset-to-arterial puncture times for both higher odds of functional independence (*p*-interaction = 0.014) and lower mortality (*p*-interaction = 0.037) at 90 days. Similarly, significant higher treatment effects were observed with longer onset-to-reperfusion times in terms of lower mortality (*p*-interaction = 0.049) at 90 days. There was no association between stroke onset-to-arterial puncture or reperfusion times and sICH in either EVT or control patients.

Notably, in the EVT group, longer times from randomization-to-reperfusion (mTICI 2b–3) were significantly associated with worse disability status (aOR per 60 minutes 0.58, 95% CI 0.37–0.91; *p* = 0.018) and lower chances of functional independence (aOR per 60 minutes 0.51, 95% CI 0.28–0.96; *p* = 0.038) at 90 days. No significant association was found between randomization-to-reperfusion times and either sICH or 90-day mortality (Table 2, Figure 2, eFigure 2–7).

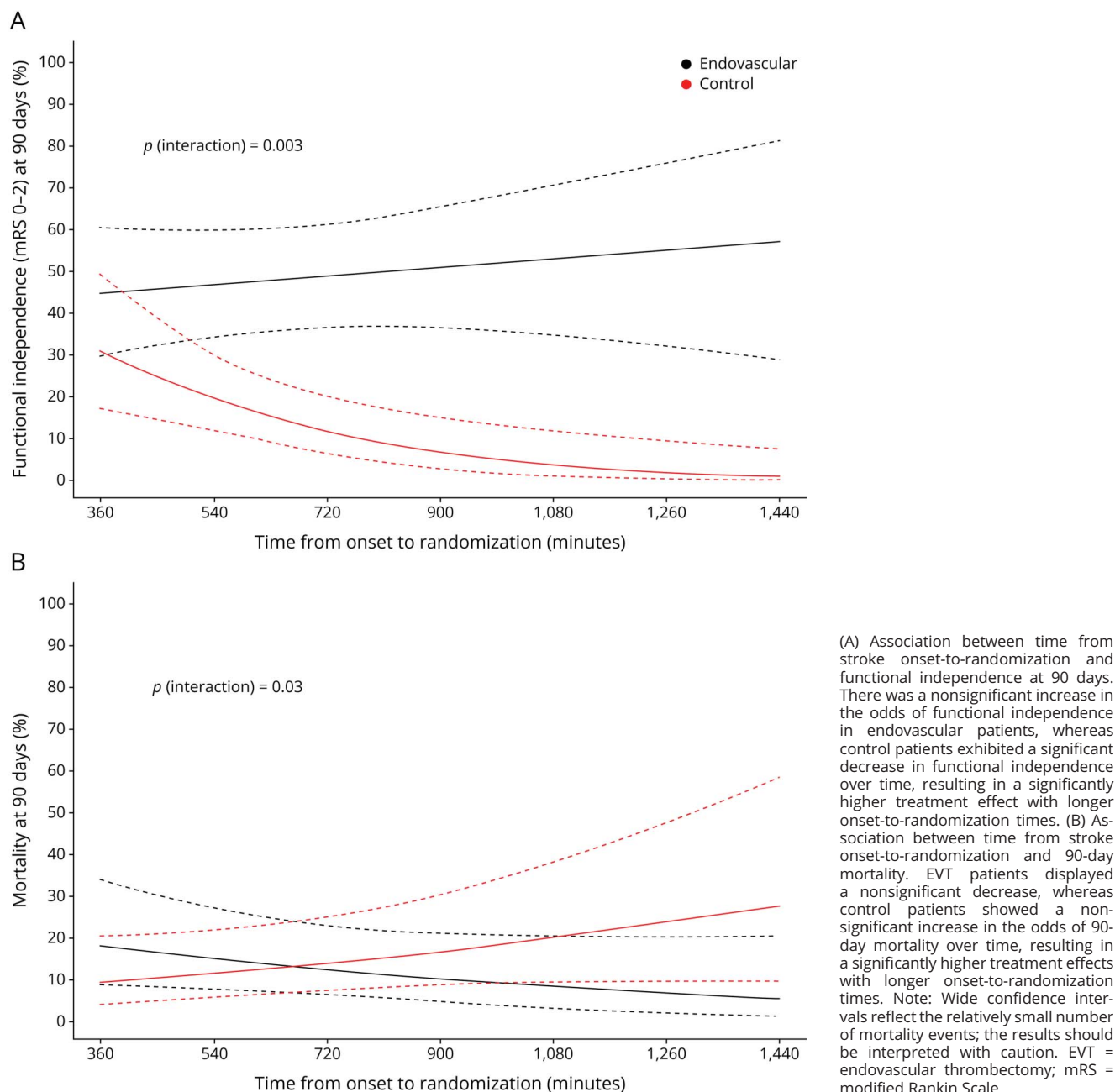
Treatment effects by onset to randomization classified into 3-time windows (6–12 hours, 12–16 hours, and 16–24 hours).

A statistically significant reduction in the overall degree of disability (mRS ordinal shift) at 90 days was observed with EVT across all 3 time epochs but with numerical differences in effect size with a maximum effect at the 12–16 hours (*n* = 120; aOR 6.96, 95% CI 3.22–15.06; *p* < 0.00001) followed by the 16–24 hours (*n* = 52; aOR 3.82 95% CI 1.07–13.69; *p* = 0.0401) and then the 6–12 hours (*n* = 332; aOR 1.76, 95% CI 1.18–2.62; *p* = 0.006) time window (*p*-interaction = 0.015) (eFigure 5).

Similarly, the rates of 90-day functional independence (mRS 0–2) were significantly higher with EVT at all time epochs but the treatment effect differed between time windows: outcomes in the 6–12-hour window were 47.5% (84/177) vs 25.8% (40/155) (aOR 2.60, 95% CI 1.63–4.14; *p* < 0.001), in the 12–16-hour window 45.9% (28/61) vs 8.5% (5/59) (aOR 9.17, 95% CI 3.19–26.36; *p* < 0.001), and in the 16–24-hour window 35.7% (10/28) vs 4.2% (1/24) (aOR 12.78, 95% CI 1.42–115.25; *p* = 0.024) (*p*-interaction < 0.001, eTable 2).

Significantly higher treatment effects were observed with longer onset-to-expected arterial puncture times in terms of both higher odds of functional independence (*p*-interaction = 0.014) and lower mortality (*p*-interaction = 0.037) at 90 days

Figure 1 Time From Stroke Onset-to-Randomization vs 90-Day Clinical Outcomes



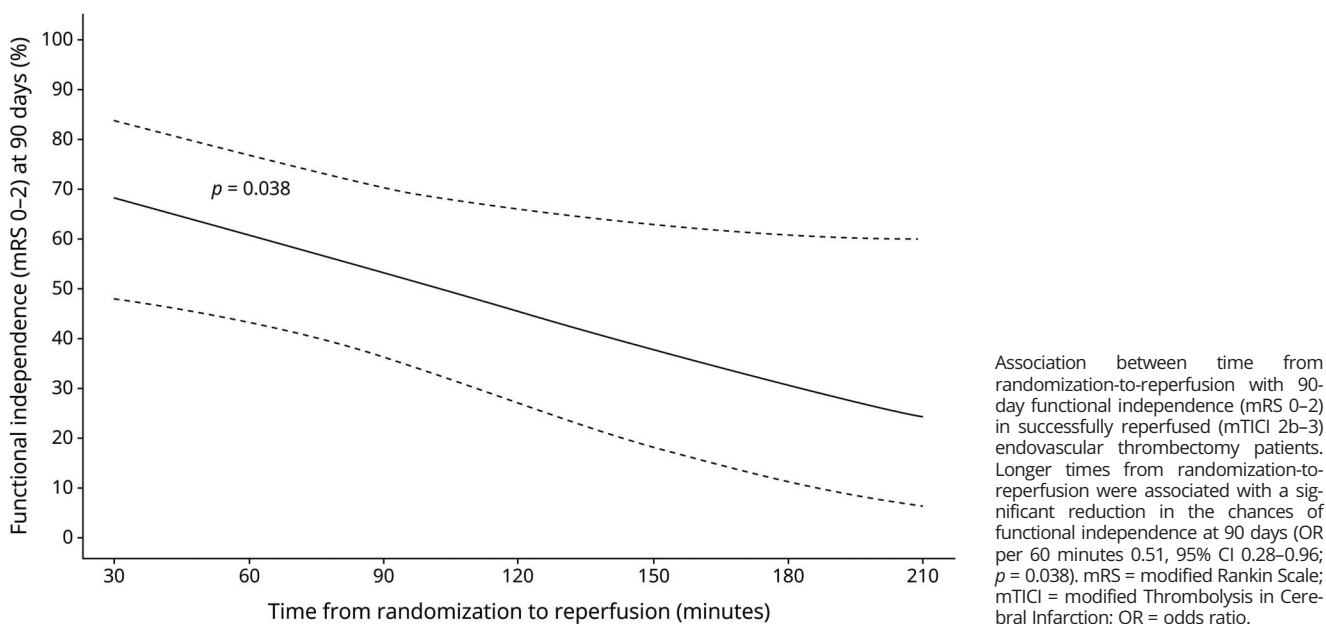
(eFigure 5, A and B). Similarly, significantly higher treatment effects were observed with longer onset-to-expected reperfusion times in terms of lower mortality (p -interaction = 0.049) at 90 days (eFigure 6, A and B). These onset-based patterns closely mirrored the onset-to-randomization findings (Figure 1, A and B), as expected given that onset-to-randomization constitutes the predominant component of both intervals. As a sensitivity analysis, randomization-to-actual arterial puncture time was not significantly associated with functional independence ($p = 0.95$) or mortality ($p = 0.24$) in patients with EVT (eFigure 7, A and B), contrasting with the significant associations observed for randomization-

to-reperfusion time (Figure 3) and suggesting that achievement of reperfusion, rather than the speed of procedural initiation, is the critical determinant of outcome. There was no association between stroke onset-to-arterial puncture or reperfusion times and sICH in either EVT or control patients.

Discussion

This analysis of the AURORA data set offers novel insights into the relationship between treatment times and the benefits of EVT in the extended (6–24 hours) time window. In

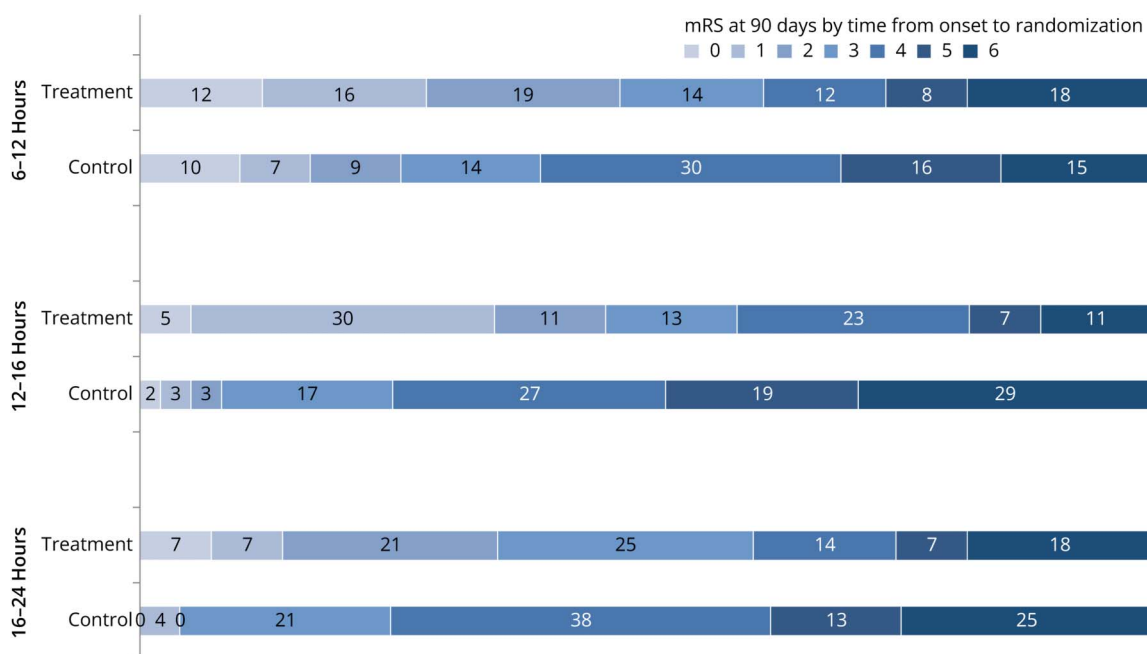
Figure 2 Impact of Randomization-to-Reperfusion Time on Functional Independence



contrast to findings from randomized trials conducted in the early time window, no significant association was observed between shorter onset-to-treatment times and improved outcomes in our cohort. Paradoxically, the treatment effect of

EVT appeared greater in the later portions of the extended window compared with earlier periods. This finding was primarily driven by a steeper decline in outcomes among medically managed patients relative to those receiving EVT.

Figure 3 Distribution of mRS Scores at 90 Days According Stratified by Onset-to-Randomization Time Windows



Distribution of mRS scores at 90 days according to treatment groups stratified by onset-to-randomization time windows. 6-12 hours: N = 332 (EVT n = 177, control n = 155), aOR 1.76 (1.18-2.62), $p = 0.006$. 12-16 hours: N = 120 (EVT n = 61, control n = 59), aOR 6.96 (3.22-15.06), $p < 0.00001$. 16-24 hours: N = 52 (EVT n = 28, control n = 24), aOR 3.82 (1.07-13.69), $p = 0.040$. Test for interaction across time windows: $p = 0.015$. aOR = adjusted odds ratio; EVT = endovascular thrombectomy; mRS = modified Rankin Scale.

Notably, within the thrombectomy group, a strong association was observed between shorter randomization-to-reperfusion times and favorable 90-day functional outcomes, with significant decays in the chances of good outcomes over time.

At first glance, these findings may imply that time to treatment is less critical for patients undergoing EVT in the extended time window. However, a fundamental limitation of studies assessing the relationship between treatment time and EVT outcomes is the systematic exclusion of patients who fail to meet imaging eligibility criteria—such as infarct size or penumbral thresholds—at presentation. This reliance on baseline imaging as a selection tool effectively resets the treatment clock to the time of presentation. The AURORA meta-analysis population reflects a highly selected subgroup with clinical-imaging mismatch: patients presenting with moderate to severe neurologic deficits yet small to moderate infarct volumes, despite delayed presentation. This profile suggests preserved collateral circulation and slow infarct evolution—so-called “slow progressors”—in whom the urgency of reperfusion may be inherently reduced.¹⁷⁻²⁰ Ideally, analytic populations would include not only patients who qualified for treatment at later time points but also those who were ineligible upon arrival but might have been candidates had they presented earlier with more favorable imaging profiles. By focusing exclusively on patients who meet specific imaging criteria, current studies introduce a denominator bias, omitting those excluded due to prehospital delays. This bias leads to a substantial underestimation of the harmful effects of time on outcomes across the broader stroke population and may inadvertently reduce the perceived urgency of rapid intervention.^{21,22}

The imaging-based selection process predictably excludes “fast progressors”—patients with time-sensitive physiology who are less likely to meet eligibility criteria as time elapses. Approximately 15% of anterior circulation LVO patients presenting within 6 hours are excluded from EVT based solely on low ASPECTS (0–5).^{23,24} In the EXTEND-IA trial, which enrolled patients eligible for intravenous alteplase within 4.5 hours and EVT within 6 hours, 25% were excluded based on perfusion imaging criteria alone.²⁵ Therefore, it is reasonable to infer that approximately 1 in 5 patients included in previously published analyses of the effect of time-to-treatment on outcomes in the early window were excluded from the analytic denominator due to imaging selection—precisely those most vulnerable to delays in treatment. Supporting this, DEFUSE 2 found that nearly 20% of patients had rapid infarct growth rates (15–100 mL/h).^{26,27}

The Highly Effective Reperfusion Evaluated in Multiple Endovascular Stroke Trials (HERMES) meta-analysis included participant-level data from 1,287 patients treated within 6 hours across 5 randomized trials. While HERMES demonstrated that the benefit of EVT declined with increasing time from symptom onset to reperfusion, closer

analysis revealed that this association was not significantly modified by time from symptom onset to emergency department (ED) arrival or randomization. Instead, the effect of time on outcomes became more apparent for intervals beginning at ED arrival, including ED arrival to randomization, arterial puncture, and reperfusion.⁴ Given the lack of biological plausibility for hours-long delays from symptom onset to ED arrival to be less impactful than minutes-long in-hospital delays, these findings illustrate a denominator fallacy: As time advances, more patients are excluded due to infarct growth beyond imaging thresholds. Their exclusion from treatment and analysis creates a selective population, biasing estimates toward a weaker association between time and treatment effect. This exclusion is even more pronounced in the late window. Approximately 25% of patients presenting between 6 and 24 hours are excluded based on ASPECTS 0–5 alone,²³ with higher exclusion rates when advanced perfusion imaging is applied. In DEFUSE 3 (6–16-hour window), 35% of screened patients did not meet imaging eligibility criteria.¹¹

Our findings are in alignment with prior studies demonstrating a lack of treatment effect modification by time to treatment among extended window patients with small to moderate infarcts. In the HERMES meta-analysis, the association between time from symptom onset to arterial puncture and EVT treatment effect lost statistical significance beyond 7.3 hours from onset. Similarly, subgroup analyses from the DAWN and DEFUSE 3 trials did not demonstrate treatment effect modification by time to treatment.^{10,28} In addition, emerging real-world evidence suggests that time to treatment may exert less influence on outcomes in this population. An analysis of 1,603 patients from the Trevo Registry compared the influence of time to treatment on 90-day outcomes in early vs late presenters.²⁹ In the early window, time to treatment was significantly associated with both overall disability (aOR for ≥ 1 -point mRS shift 0.75, 95% CI 0.66–0.86; $p < 0.001$) and functional independence (aOR for mRS 0–2: 0.73, 95% CI 0.62–0.86; $p < 0.001$). By contrast, no significant associations were observed in the extended window (0.96, 95% CI 0.90–1.02; $p = 0.15$ and 0.97, 95% CI 0.90–1.04; $p = 0.41$, respectively). Similarly, an analysis from the Get With The Guidelines–Stroke registry, including 6,756 patients, demonstrated a nonlinear relationship between time and outcome, with steeper declines in benefit between 30 to 270 minutes than between 271 to 480 minutes.³⁰ Therefore, the collective data suggest that the effect of time to EVT on functional outcomes is not uniform but rather time dependent—exerting a stronger influence in the early hours after stroke onset, followed by a more attenuated effect with later presentation. This observation supports continued efforts to extend the EVT window beyond 24 hours in properly selected patients.³¹

However, a critical finding of this study was that longer time from randomization to reperfusion was significantly associated with worse functional outcomes (cOR for mRS shift per 60 minutes 0.58, 95% CI 0.37–0.91; $p = 0.018$) and lower

odds of functional independence (aOR for mRS 0–2 per 60 minutes 0.51, 95% CI 0.28–0.96; $p = 0.038$) at 90 days. These findings highlight that the apparent dissociation between time and outcomes in the extended window is largely artificial—driven by the imaging-based reset of the physiologic clock to the time of presentation, which masks the consequences of prehospital delays. Even in predominantly “slow-progressor” populations, time to treatment remains a critical determinant of outcome.

Our findings should be interpreted in the context of several limitations inherent to post hoc analyses. First, differences in study criteria and population characteristics across trials may introduce bias. However, random-effects models were used to account for between-study variability and potential confounding. Second, 2 recently published extended window randomized trials were not included due to their focus on distinct populations. The MR CLEAN-LATE trial enrolled patients with LVO within 6–24 hours from symptom onset, using collateral flow on CT angiography ($\geq$ grade 1) as the selection criterion. Patients with estimated ischemic core volume <25 mL on CT perfusion were excluded—a subgroup that would have overlapped with patients eligible for DAWN and DEFUSE 3 based on this parameter—although this represents only 1 component of those trials’ multifactorial eligibility criteria, limiting direct comparison with the present cohort.³² Patients meeting the clinical and perfusion imaging criteria from DAWN or DEFUSE 3 were specifically excluded, limiting overlap with the present cohort. The RESILIENT-Extend trial randomized patients with LVO in the 8- to 24-hour window using age-adjusted NIHSS–ASPECTS mismatch criteria.³³ Conducted in the Brazilian public health system, it included a more socioeconomically vulnerable population and reported 90-day mortality in the EVT group that was 2- to 3-fold higher than in ESCAPE and DEFUSE 3. Notably, while previous studies showed numerically greater benefit in older patients, RESILIENT-Extend was the first to suggest potential harm of EVT in this subgroup. Third, definitions of sICH were based on trial-specific criteria, introducing variability in outcome assessment. Fourth, reliance on “time last known well” (TLKW) as a surrogate for symptom onset introduces substantial measurement error, particularly in the extended window where unwitnessed and wake-up strokes are common. As presented in Table 1, onset mode shifts markedly across time epochs, with unwitnessed strokes predominating in later windows—most notably in the 16–24-hour interval (DAWN trial patients only). A witnessed-onset-restricted analysis was not feasible due to incomplete data across trials. However, DAWN trial findings using time since first observed and onset mode indicate that TLKW-based timing substantially overestimates true ischemia duration, especially for wake-up strokes, which often behave as fast progressors despite delayed presentation. This imprecision likely attenuates associations between onset-to-treatment intervals and outcomes. Supporting this, DAWN analyses suggest that wake-up strokes frequently occur shortly before awakening rather than at TLKW, reinforcing

misclassification concerns. Importantly, wake-up patients in DAWN exhibited a significant decline in functional independence with longer time to reperfusion ($p = 0.018$), whereas witnessed and unwitnessed strokes showed no such dependency—underscoring heterogeneity by onset mode and the limitations of TLKW as a proxy for true onset. Collectively, these findings highlight that imaging-based selection effectively “resets” the physiologic clock, masking the detrimental impact of prehospital delays in extended-window trials.³⁴ Fifth, the declining use of intravenous thrombolysis with later presentation (6–12 hours: 18.9%; 12–16 hours: 2.5%; 16–24 hours: 0%; Table 1) represents a potential confounder, as thrombolysis-treated control patients may have had better outcomes independent of time effects. However, the small numbers of thrombolysis-treated patients in the extended window precluded robust adjustment for this variable. Although the role of EVT is expanding—particularly in patients with large infarcts or posterior circulation strokes—time sensitivity in these populations remains to be defined.^{35–37} Nevertheless, the multinational nature of this data set supports the external validity and broad applicability of the findings.

In conclusion, in the AURORA meta-analysis, predominantly selected with advanced imaging and enriched with slow progressors, the treatment effect of thrombectomy appeared to paradoxically increase with longer onset-to-presentation times—underscoring the validity of selection based on physiologic information. However, this should not diminish the importance of time, as strict imaging selection excludes many patients and artificially masks the harmful impact of preimaging delays. In fact, longer randomization-to-reperfusion intervals were associated with worse outcomes, reinforcing the principle that “time is brain,”³ even in highly selected populations. Future studies should evaluate time effects in extended-window patients selected with simpler imaging and in those with large infarcts or posterior circulation strokes.

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Author Contributions

R.G. Nogueira: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. T.G. Jovin: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M.G. Lansberg: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study

concept or design. D.C. Haussen: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A.P. Jadhav: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S. Brown: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M. Ribo: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S.C. Martins: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. F.J. Mont'Alverne: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M.D. Hill: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M. Goyal: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M.F. Doheim: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. J.D. Mocco: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. D.S. Liebeskind: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. J.J. Heit: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M.P. Marks: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. G.W. Albers: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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Disclosure

R.G. Nogueira has served as a paid consultant or advisory board member for Anaconda, Biogen, Boehringer Ingelheim, Cerenovus, Genentech, Hybernia, Hyperfine, Imperative Care, Medtronic, Phenox, Philips, Prolong Pharmaceuticals, Shanghai Wallaby, Stryker Neurovascular, Synchron, and Takeda Pharmaceutical. He holds stock options received through advisory roles in Astrocyte, Brainomix, Cerebrotech, Ceretrieve, Corindus Vascular Robotics, CrestecBio Inc., Euphrates Vascular Inc., Perfuze, RapidPulse, Sensitive, Vesalio, and Viz-AI. He additionally holds equity interests in Brain4Care, Cerebrotech, Perfuze, Piraeus Medical, Quantanosis AI, Reist/Q'Apel Medical, Truvic, Tulavi Therapeutics, Unity Medical, Vastrax, Viseon, and Viz-AI. He serves as co-Principal Investigator of the ENDOLOW trial (funded by Cerenovus to Emory University) and as Principal Investigator of the DUSK trial (funded

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References

- Goyal M, Menon BK, Van Zwam WH, et al. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five

- randomised trials. *Lancet*. 2016;387(10029):1723-1731. doi:10.1016/S0140-6736(16)00163-X
2. Jovin TG, Nogueira RG, Lansberg MG, et al. Thrombectomy for anterior circulation stroke beyond 6 h from time last known well (AURORA): a systematic review and individual patient data meta-analysis. *Lancet*. 2022;399(10321):249-258. doi:10.1016/S0140-6736(21)01341-6
3. Saver JL. Time is brain: quantified. *Stroke*. 2006;37(1):263-266. doi:10.1161/01.STR.0000196957.55928.ab
4. Saver JL, Goyal M, van der Lugt A, et al. Time to treatment with endovascular thrombectomy and outcomes from ischemic stroke: a meta-analysis. *JAMA*. 2016;316(12):1279-1288. doi:10.1001/jama.2016.13647
5. Franssen PS, Berkhemer OA, Lingsma HF, et al. Time to reperfusion and treatment effect for acute ischemic stroke: a randomized clinical trial. *JAMA Neurol*. 2016;73(2):190-196. doi:10.1001/jamaneurol.2015.3886
6. Menon BK, Sajobi TT, Zhang Y, et al. Analysis of workflow and time to treatment on thrombectomy outcome in the endovascular treatment for small core and proximal occlusion ischemic stroke (ESCAPE) randomized, controlled trial. *Circulation*. 2016;133(23):2279-2286. doi:10.1161/CIRCULATIONAHA.115.019983
7. Ribo M, Molina CA, Cobo E, et al. Association between time to reperfusion and outcome is primarily driven by the time from imaging to reperfusion. *Stroke*. 2016;47(4):999-1004. doi:10.1161/STROKEAHA.115.011721
8. Goyal M, Jadhav AP, Bonafe A, et al. Analysis of workflow and time to treatment and the effects on outcome in endovascular treatment of acute ischemic stroke: results from the SWIFT PRIME randomized controlled trial. *Radiology*. 2016;279(3):888-897. doi:10.1148/radiol.2016160204
9. Sheth SA, Jahan R, Gralla J, et al. Time to endovascular reperfusion and degree of disability in acute stroke. *Ann Neurol*. 2015;78(4):584-593. doi:10.1002/ana.24474
10. Nogueira RG, Jadhav AP, Haussen DC, et al. Thrombectomy 6 to 24 hours after stroke with a mismatch between deficit and infarct. *N Engl J Med*. 2018;378(1):11-21. doi:10.1056/NEJMoa1706442
11. Albers GW, Marks MP, Kemp S, et al. Thrombectomy for stroke at 6 to 16 hours with selection by perfusion imaging. *N Engl J Med*. 2018;378(8):708-718. doi:10.1056/NEJMoa1713973
12. Goyal M, Demchuk AM, Menon BK, et al. Randomized assessment of rapid endovascular treatment of ischemic stroke. *N Engl J Med*. 2015;372(11):1019-1030. doi:10.1056/NEJMoa1414905
13. Martins SO, Mont'Alverne F, Rebello LC, et al. Thrombectomy for stroke in the public health care system of Brazil. *N Engl J Med*. 2020;382(24):2316-2326. doi:10.1056/NEJMoa2000120
14. Mocco J, Siddiqui AH, Fiorella D, et al. POSITIVE: perfusion imaging selection of ischemic stroke patients for endovascular therapy. *J Neurointerv Surg*. 2022;14(2):126-132. doi:10.1136/neurintsurg-2021-017315
15. Jovin TG, Chamorro A, Cobo E, et al. Thrombectomy within 8 hours after symptom onset in ischemic stroke. *N Engl J Med*. 2015;372(24):2296-2306. doi:10.1056/NEJMoa1503780
16. Zaidat OO, Yoo AJ, Khatri P, et al. Recommendations on angiographic revascularization grading standards for acute ischemic stroke: a consensus statement. *Stroke*. 2013;44(9):2650-2663. doi:10.1161/STROKEAHA.113.001972
17. Rocha M, Jovin TG. Fast versus slow progressors of infarct growth in large vessel occlusion stroke: clinical and research implications. *Stroke*. 2017;48(9):2621-2627. doi:10.1161/STROKEAHA.117.017673
18. Liebeskind DS, Saber H, Xiang B, et al. Collateral circulation in thrombectomy for stroke after 6 to 24 hours in the DAWN trial. *Stroke*. 2022;53(3):742-748. doi:10.1161/STROKEAHA.121.034471
19. Mohammad MH, Haussen DC, Pisani L, et al. Characterizing fast and slow progressors in anterior circulation large vessel occlusion strokes. *Interv Neuroradiol*. 2022;29(4):379-385. doi:10.1177/15910199221083100
20. Nguyen TN, Abdalkader M, Nagel S, et al. Noncontrast computed tomography vs computed tomography perfusion or magnetic resonance imaging selection in late presentation of stroke with large-vessel occlusion. *JAMA Neurol*. 2022;79(1):22-31. doi:10.1001/jamaneurol.2021.4082
21. Goyal M, Jadhav AP. Denominator fallacy revisited. *J Neurointerv Surg*. 2017;9(10):915-916. doi:10.1136/neurintsurg-2016-012486
22. Nogueira RG, Ribo M. Endovascular treatment of acute stroke. *Stroke*. 2019;50(9):2612-2618. doi:10.1161/STROKEAHA.119.023811
23. Mokin M, Pendurthi A, Ljubimov V, et al. ASPECTS, large vessel occlusion, and time of symptom onset: estimation of eligibility for endovascular therapy. *Neurosurgery*. 2018;83(1):122-127. doi:10.1093/neuros/nyx352
24. Requena M, Perez de la Ossa N, Abilleira S, et al. Predictors of endovascular treatment among stroke codes activated within 6 hours from symptom onset. *Stroke*. 2018;49(9):2116-2121. doi:10.1161/STROKEAHA.118.021316
25. Campbell BC, Mitchell PJ, Kleinig TJ, et al. Endovascular therapy for ischemic stroke with perfusion-imaging selection. *N Engl J Med*. 2015;372(11):1009-1018. doi:10.1056/NEJMoa1414792
26. Wheeler HM, Mlynash M, Inoue M, et al. The growth rate of early DWI lesions is highly variable and associated with penumbral salvage and clinical outcomes following endovascular reperfusion. *Int J Stroke*. 2015;10(5):723-729. doi:10.1111/ijis.12436
27. Albers GW. Late window paradox. *Stroke*. 2018;49(3):768-771. doi:10.1161/STROKEAHA.117.020200
28. Lansberg MG, Mlynash M, Hamilton S, et al. Association of thrombectomy with stroke outcomes among patient subgroups: secondary analyses of the DEFUSE 3 randomized clinical trial. *JAMA Neurol*. 2019;76(4):447-453. doi:10.1001/jamaneurol.2018.4587
29. Nogueira RG, Jovin TG, Haussen DC, et al. Influence of time to endovascular stroke treatment on outcomes in the early versus extended window paradigms. *Int J Stroke*. 2022;17(3):331-340. doi:10.1177/17474930211006304
30. Jahan R, Saver JL, Schwamm LH, et al. Association between time to treatment with endovascular reperfusion therapy and outcomes in patients with acute ischemic stroke treated in clinical practice. *JAMA*. 2019;322(3):252-263. doi:10.1001/jama.2019.8286
31. Sarraj A, Kleinig TJ, Hassan AE, et al. Association of endovascular thrombectomy vs medical management with functional and safety outcomes in patients treated beyond 24 hours of last known well: the SELECT late study. *JAMA Neurol*. 2023;80(2):172-182. doi:10.1001/jamaneurol.2022.4714
32. Olthuis SGH, Pirson FAV, Pinckaers FME, et al. Endovascular treatment versus no endovascular treatment after 6-24 h in patients with ischaemic stroke and collateral flow on CT angiography (MR CLEAN-LATE) in the Netherlands: a multicentre, open-label, blinded-endpoint, randomised, controlled, phase 3 trial. *Lancet*. 2023;401(10385):1371-1380. doi:10.1016/S0140-6736(23)00575-5
33. Nogueira R, Mont'Alverne F, Cabral L, et al. Randomization of endovascular treatment with stent-retriever and/or thromboaspiration vs. best medical therapy in acute ischemic stroke due to large vessel occlusion trial in the extended time window (RESILIENT-Extend): a trial of extended window thrombectomy without advanced imaging in a developing country. Presented at: International Stroke Conference; Phoenix, AZ; 2024.
34. Nogueira RG, Doheim MF, Jadhav AP, et al. Mode of onset modifies the effect of time to endovascular reperfusion on clinical outcomes after acute ischemic stroke: an analysis of the DAWN trial. *Ann Neurol*. 2024;96(2):356-364. doi:10.1002/ana.26968
35. Doheim MF, Hagrass AI, Elrefaey M, et al. From therapeutic nihilism to armamentarium: a meta-analysis of randomized clinical trials assessing safety and efficacy of endovascular therapy for acute large ischemic strokes. *Interv Neuroradiol*. 2023. doi:10.1177/15910199231170681
36. Jovin TG, Li C, Wu L, et al. Trial of thrombectomy 6 to 24 hours after stroke due to basilar-artery occlusion. *N Engl J Med*. 2022;387(15):1373-1384. doi:10.1056/NEJMoa2207576
37. Tao C, Nogueira RG, Zhu Y, et al. Trial of endovascular treatment of acute basilar-artery occlusion. *N Engl J Med*. 2022;387(15):1361-1372. doi:10.1056/NEJMoa2206317